Application of Generalized Linear Model for Binary Response on the “Evans” Dataset

Wilson Yao
Dongxuan Li
Matthew Kim
Seth Paine

Introduction

Our project focuses on applying logistic regression, a fundamental statistical technique, to a real-world dataset

- The logistic regression model is a form of a generalized linear model (GLM)
- GLMs are an extension of the linear model to allow for a range of distributions that the response can follow, including binomial for the case of binary responses

In this project we aim to use the logistic regression model to identify predictors of a significant worldwide public health concern, coronary heart disease.

Dataset

The data set ”Evans” that is available in R-package “lbreg”.
Description: Data from a cohort study in which white males in Evans County were followed for 7 years, with coronary heart disease as the outcome of interest.

A data frame with 609 observations on the following 9 variables.

- CDH: outcome variable; 1 = coronary heart disease
- CAT: 1 = high, 0 = normal; catecholamine level
- AGE: age (in years)
- CHL: cholesterol, mg/dl
- SMK: 1 = subject has ever smoked
- ECG: 1 = presence of electrocardiogram abnormality
- DBP: diastolic blood pressure, mmHg
- SBP: systolic blood pressure, mmHg
- HPT: 1 = SBP greater than or equal to 160 or DBP greater than or equal to 95

Analysis

$$H_0: \beta_{CAT} = \beta_{AGE} = \beta_{CHL} = \beta_{SMK} = \beta_{ECG} = \beta_{DBP} = \beta_{HSBP} = \beta_{HPT} = 0,$$

$$H_a: \text{at least one of } \beta_{CAT}, \beta_{AGE}, \beta_{CHL}, \beta_{SMK}, \beta_{ECG}, \beta_{DBP}, \beta_{HSBP}, \beta_{HPT} \neq 0.$$

Firstly, we fit our full model using a logistic regression model:

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.764709   1.579763  -4.282 1.85e-05 ***
CAT1         0.750257   0.377231   1.989  0.04672 *
AGE          0.037781   0.016031   2.357  0.01844 *
CHL          0.008674   0.003296   2.632  0.00850 **
SMK1         0.867474   0.307218   2.824  0.00475 **
ECG1         0.403426   0.296125   1.362  0.17309
DBP          0.012402   0.014284   0.868  0.38526
SBP         -0.010571   0.008073  -1.309  0.19038
HPT1         0.532752   0.383747   1.388  0.16505
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Checking for over-dispersion using a quasibinomial GLM:

```
Call:
glm(formula = CDH ~ ., family = quasibinomial, data = Evans)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -6.764709   1.581990  -4.276 2.21e-05 ***
CAT1         0.750257   0.377763   1.986  0.04748 *
AGE          0.037781   0.016054   2.353  0.01892 *
CHL          0.008674   0.003301   2.628  0.00881 **
SMK1         0.867474   0.307651   2.820  0.00497 **
ECG1         0.403426   0.296543   1.360  0.17420
DBP          0.012402   0.014304   0.867  0.38628
SBP         -0.010571   0.008084  -1.308  0.19150
HPT1         0.532752   0.384288   1.386  0.16616
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 1.002821)
```

Our dispersion parameter is very close to 1, suggesting that we do not have an over-dispersion problem

Performing stepwise selection using AIC criterion:

```
Start:  AIC=416.6
CDH ~ CAT + AGE + CHL + SMK + ECG + DBP + SBP + HPT

       Df Deviance    AIC
- DBP   1   399.35 415.35
- SBP   1   400.35 416.35
- ECG   1   400.41 416.41
- HPT   1   400.54 416.54
<none>      398.60 416.60
- CAT   1   402.53 418.53
- AGE   1   404.11 420.11
- CHL   1   405.44 421.44
- SMK   1   407.40 423.40
```

```
Step:  AIC=413.95
CDH ~ CAT + AGE + CHL + SMK + HPT

       Df Deviance    AIC
<none>      401.95 413.95
+ ECG   1   400.39 414.39
- HPT   1   404.66 414.66
+ SBP   1   401.31 415.31
+ DBP   1   401.81 415.81
- CAT   1   406.33 416.33
- AGE   1   406.52 416.52
- CHL   1   408.86 418.86
- SMK   1   409.65 419.65
```

Thus our important predictors are HPT, CAT, AGE, CHL, and SMK.
We drop our unimportant predictors ECG, SBP, and DBP

Analysis of our reduced model:

```
Call:
glm(formula = CDH ~ CAT + AGE + CHL + SMK + HPT, family = binomial,
    data = Evans)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.0762  -0.5297  -0.3995  -0.3048   2.5356

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.680112   1.136363  -5.879 4.14e-09 ***
CAT1         0.715810   0.340180   2.104  0.03536 *
AGE          0.032770   0.015197   2.156  0.03105 *
CHL          0.008608   0.003259   2.641  0.00827 **
SMK1         0.802906   0.303001   2.650  0.00805 **
HPT1         0.476272   0.289296   1.646  0.09970 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Since the Wald-type test is not the best choice for categorical predictors, we use a chi-square test to identify the significance of our predictors

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: CDH

Terms added sequentially (first to last)


     Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                   608     438.56
CAT   1  14.1312       607     424.43 0.0001705 ***
AGE   1   4.0483       606     420.38 0.0442171 *
CHL   1   8.2166       605     412.16 0.0041510 **
SMK   1   7.5056       604     404.66 0.0061506 **
HPT   1   2.7097       603     401.95 0.0997410 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

We identify CAT, AGE, CHL, and SMK as significant predictors at a 5% significance level.

To test whether we should use our reduced model, we compare nested models by comparing the difference in deviance to a chi-square distribution with 3 degrees of freedom

- Under the null hypothesis (no difference in models), the difference in deviance will asymptotically follow a chi-square distribution

```
Analysis of Deviance Table

Model 1: CDH ~ CAT + AGE + CHL + SMK + HPT
Model 2: CDH ~ CAT + AGE + CHL + SMK + ECG + DBP + SBP + HPT
  Resid. Df Resid. Dev Df Deviance
1       603     401.95
2       600     398.60  3   3.3468
```

Using a chi-square test, we find a p-value of 0.3412 and conclude that there is not a significant difference between the two models. Thus, we select the smaller model (model 1).

Conclusion

We have identified some significant predictors of coronary heart disease (CDH):

- Catecholamine level (CAT)
- Age
- Cholesterol (CHL)
- Smoking (SMK)

References

- Andrade BB (2019). _lbreg: Log-Binomial Regression with Constrained Optimization_. R package version 1.3.
  <https://CRAN.R-project.org/package=lbreg>.
  https://cran.r-project.org/web/packages/lbreg/lbreg.pdf

- D. Kleinbaum and M. Klein (2010) Survival Analysis: A Self-Learning Text. 3rd ed. Springer.
  http://web1.sph.emory.edu/dkleinb/logreg3.htm#data